SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	A. Kullai Reddy	Reg. No.:	192412647
Section / Batch:	ECE	Date:	15-07-2026

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct

I A. Kullai Reddy (192412647) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______

A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the?

Hexadecimal color code Analysis :-

Aim:- To compare two hexadecimal colour codes and obeserve now changing one value affects the displayed colour

Example :

FF0000 ⟶ Bright Red

F00000 ⟶ Darker Red.

Binary Representation :-

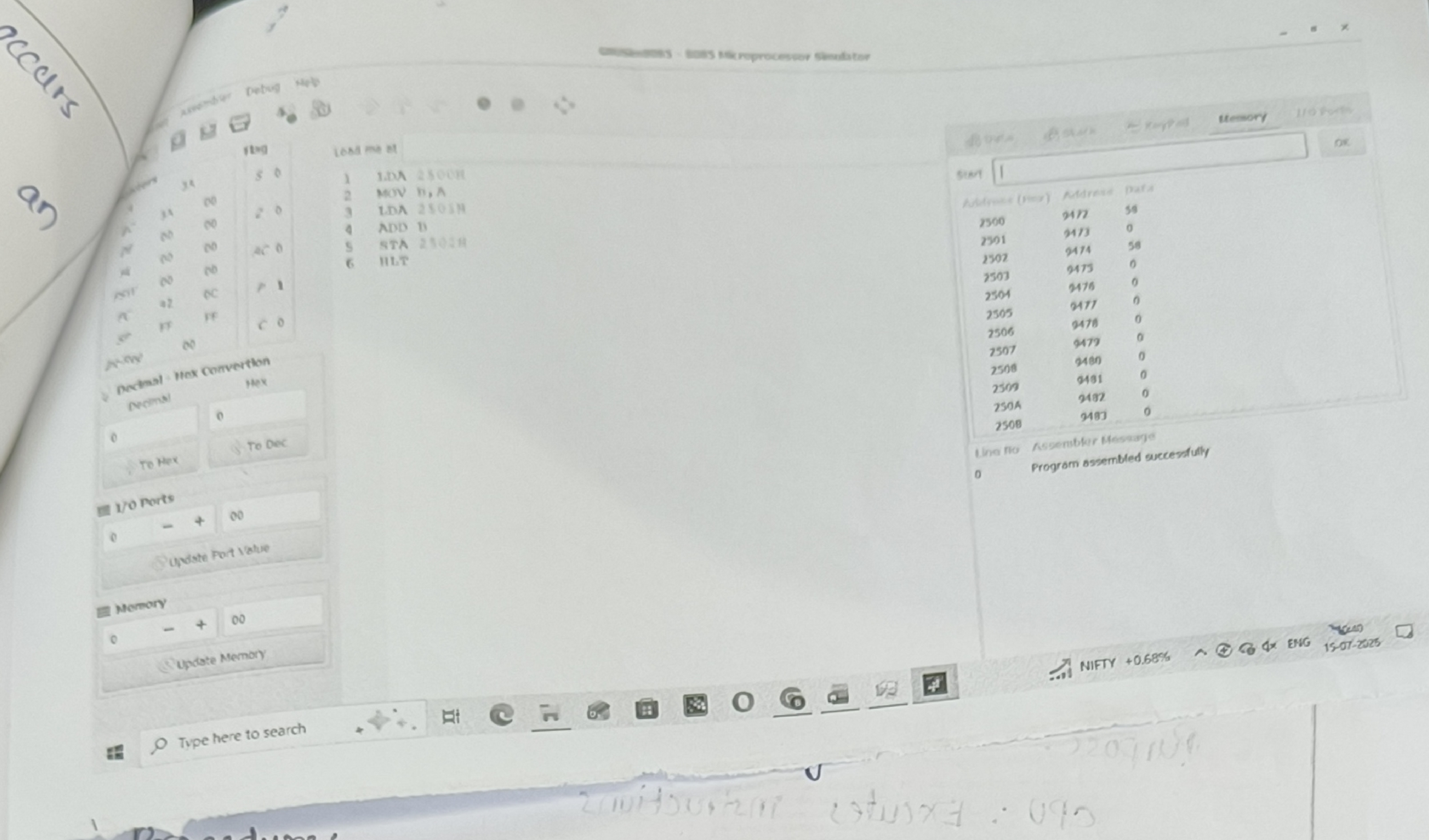
FF = 11111111

F0 = 11110000

00 = 00000000.

Obeservation :-

The colour #FF0000 has the maximum red intensity (255). changing FF to F0 reduces the red value to 240, making the colour slightly darker.



Procedure:

Open a document editor, browse the internet, and play audio simultaneously. obeserve the system performance.

Observation:

The computer becomes slower because multiple applications are running at the same time. The CPU usage increase, while I/O devices continously transfer data.

Carry (CY) : set to 1 if carry or borrow occurs

Parity (P) : set to 1 if the result contains an even number of 1s.

Flag Register Analysis :-

```
LXI H, 2500H
MOV A, M
INX H
ADD M
STA 2502 H
HLT
```

Example Data :-

2500H = 25H

2501 H = 15H

Obeserve :-

sign (S)

zero (Z)

carry (CY)

Parity (P)

conclusion :-

The flag register reflects the result of arthemetic operations. Different input values generate different flag values, which influence the execution of the program.

2) A Proggammer notices that a program produces different outputs when different input values are used. ?

Aim:-

To Obeserve the effect of different input values on the sign, zero, carry and parity flags, in the 8085 microprocessor.

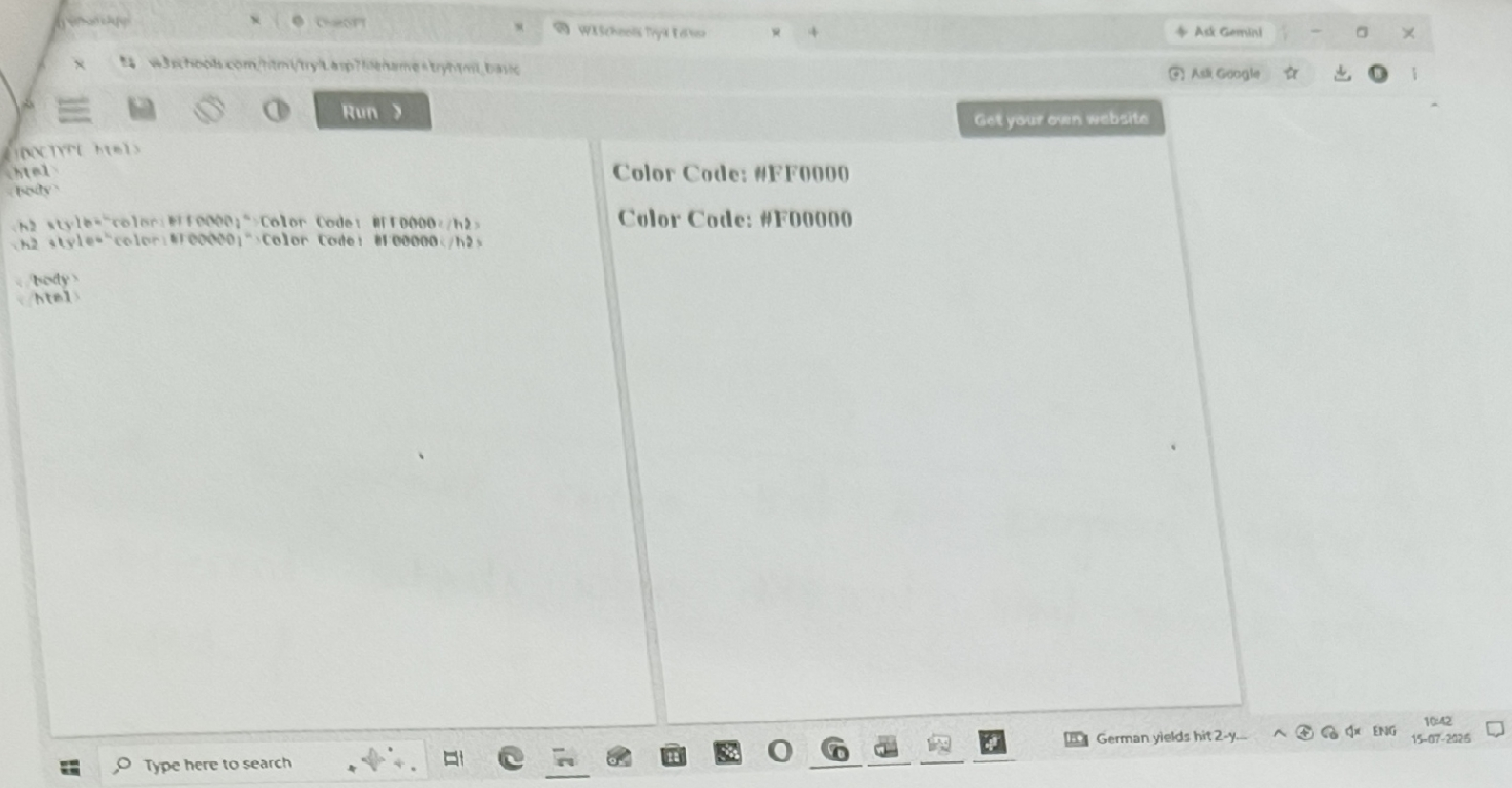
Procedure:-

Excute an 8085 program that performs arthemetic operations such as addition or subtraction using different input values. After each operation obeserve the flag register.

Obeserration:-

Sign (S) : set to 1 if the result is negative (MSB = 1)

Zero (Z) : set to 1 if the result is zero.



performs arithmetic operations such as addition or subtraction using different input values. After each operation obeserve the flag register.

Obeservation:-

Sign (S) : set to 1 if the result is negative (MSB = 1)

Zero (Z) : set to 1 if the result is zero.

Program:-

```
MVI A, FFH ;
STA 2500 H ;
MVI A, F0H ;
STA 2501 H ;

HLT
```

Purpose:-

Demonstrates storing hexadecimal values (FFH and F0H), similar to hexadecimal colour components.

Conclusion:

Changing a hexadecimal digit changes the binary representation, which changes the RGB intensity and produces a different color.

A user notices that a computer responds slowly while editing a document, browsing the internet and playing audio simultaneously.

Aim:

To study how the CPU, memory, and I/O devices work together while performing multiple tasks

Procedure:

Open a document editor, browse the Internet, and play audio simultaneously. obeserve the system Performance.

Observation:

The computer becomes slower because multiple applications are running at the same time. The CPU usage increase, while I/O devices continously transfer data.

Program:

CPU, Memory . and I/O Device Analysis.

```
MVI  A, 55H   ;
STA  2500H    ;
LDA  2500H    ;
OUT  01H      ;
HLT
```

Purpose:

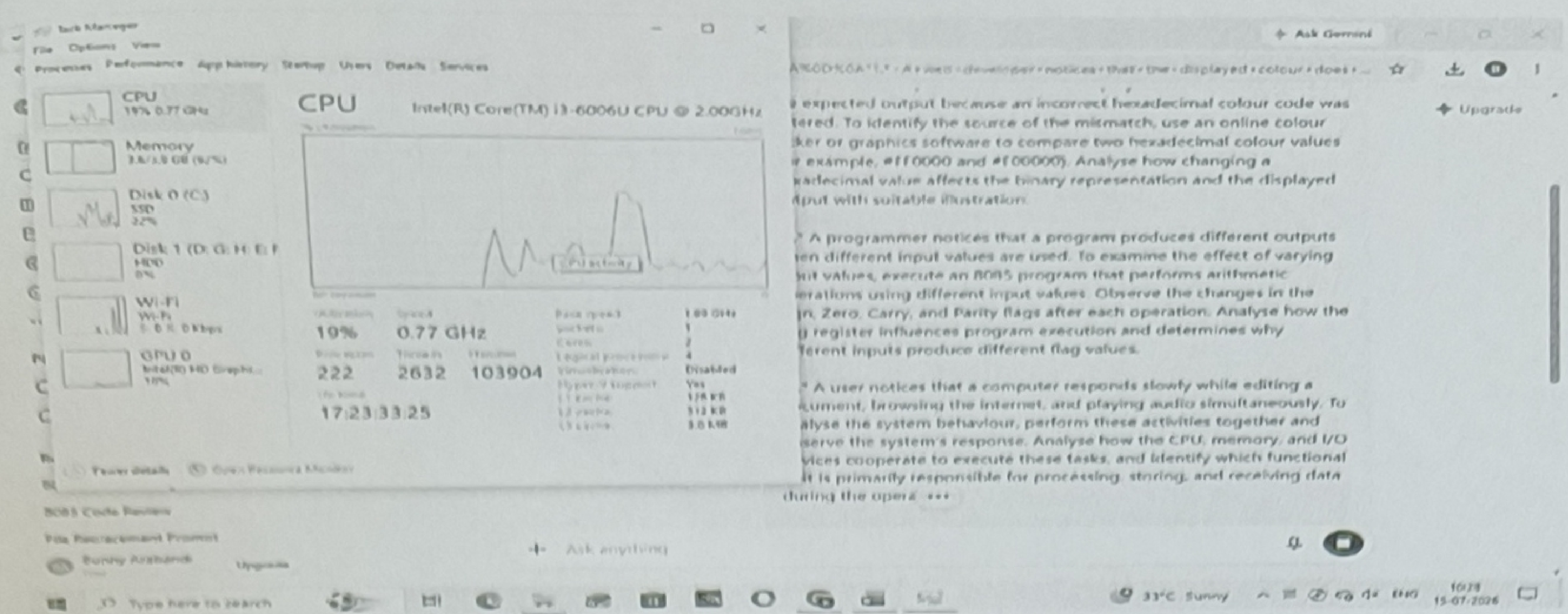
CPU : Excutes instructions

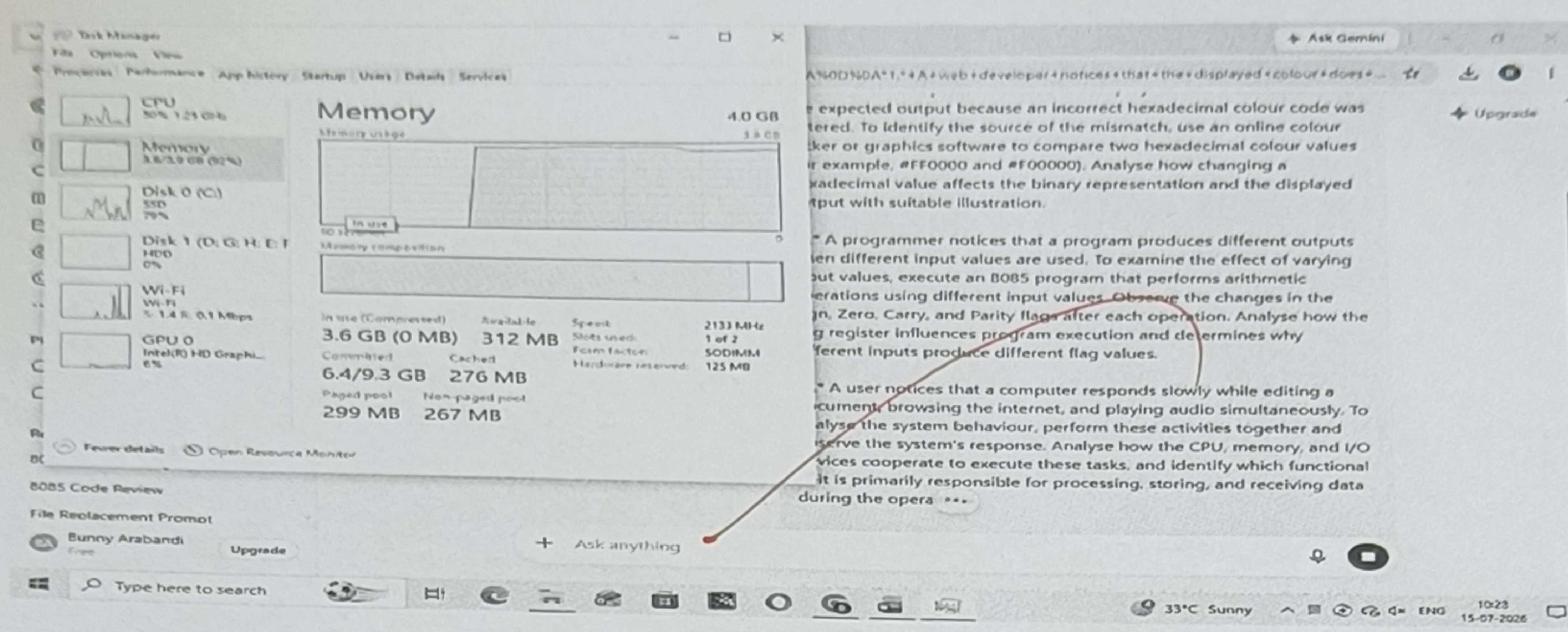
Memory: stores data at address 2500H

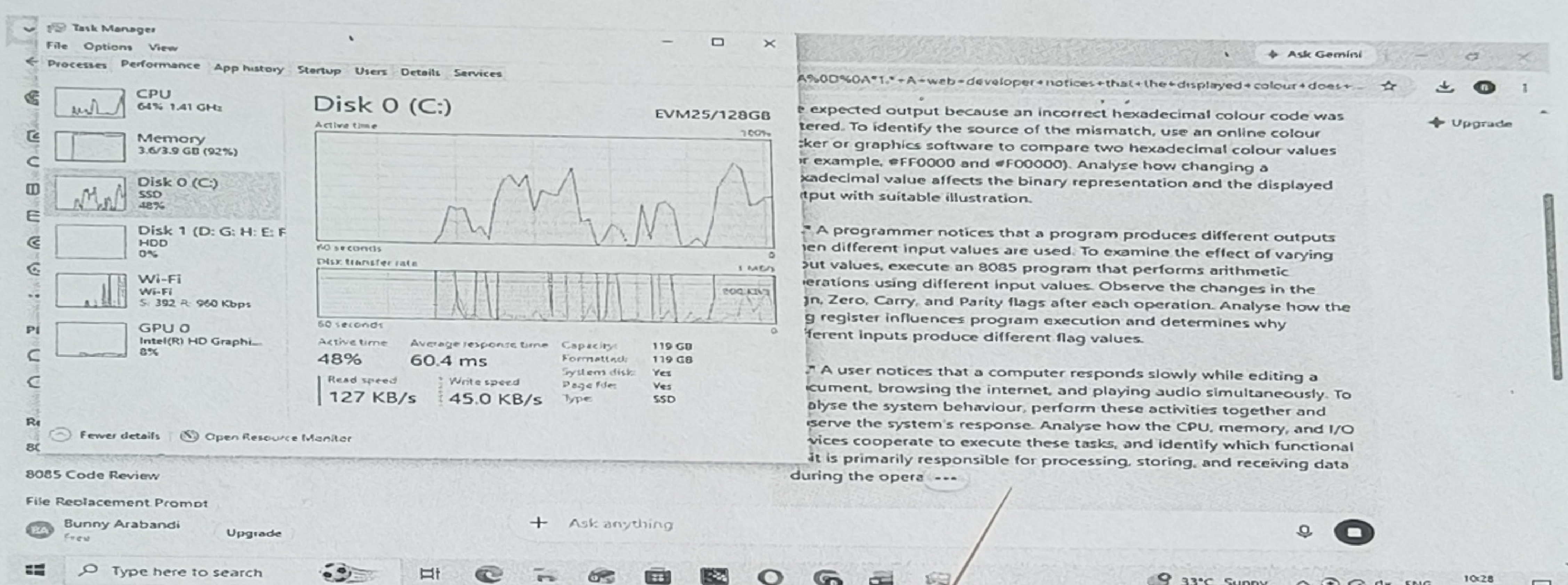
I/O : sends data to output port using the OUT instruction.

Conclusion:

The CPU, memory, and I/O devices work together to excute multiple tasks efficently. The CPU Performs processing, RAM stores temporary data, and I/O devices manage data transfer between the user and the computer.


Task Manager
CPU
Intel(R) Core(TM) i3-6006U CPU @ 2.00GHz
19%
0.77 GHz
222
2632
103904
17:23:33:25
8085 Code Review
File Replacement Prompt
Bunny Arabandi
Upgrade
Ask anything
Type here to search
33°C Sunny
15-07-2026
expected output because an incorrect hexadecimal colour code was
tered. To identify the source of the mismatch, use an online colour
ker or graphics software to compare two hexadecimal colour values
example, #FF0000 and #F00000). Analyse how changing a
adecimal value affects the binary representation and the displayed
tput with suitable illustration.
A programmer notices that a program produces different outputs
en different input values are used. To examine the effect of varying
ut values, execute an 8085 program that performs arithmetic
erations using different input values. Observe the changes in the
n, Zero, Carry, and Parity flags after each operation. Analyse how the
register influences program execution and determines why
ferent inputs produce different flag values.
A user notices that a computer responds slowly while editing a
cument, browsing the internet, and playing audio simultaneously. To
alyse the system behaviour, perform these activities together and
serve the system's response. Analyse how the CPU, memory, and I/O
vices cooperate to execute these tasks, and identify which functional
it is primarily responsible for processing, storing, and receiving data
during the opera


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Memory
4.0 GB
3.6 GB (0 MB)
312 MB
6.4/9.3 GB
276 MB
299 MB
267 MB
2133 MHz
1 of 2
SODIMM
125 MB
Fewer details
Open Resource Monitor
8085 Code Review
File Replacement Prompt
Bunny Arabandi
Upgrade
Ask anything
Type here to search
33°C Sunny
ENG
10:23
15-07-2026
Ask Gemini


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
CPU
64% 1.41 GHz
Memory
3.6/3.9 GB (92%)
Disk 0 (C:)
SSD
48%
Disk 1 (D: G: H: E: F
HDD
0%
Wi-Fi
GPU 0
Intel(R) HD Graphi...
8%
Disk 0 (C:)
EVM25/128GB
Active time
48%
Average response time
60.4 ms
Read speed
127 KB/s
Write speed
45.0 KB/s
Capacity: 119 GB
Formatted: 119 GB
System disk: Yes
Page file: Yes
Type: SSD
Fewer details
Open Resource Monitor
8085 Code Review
File Replacement Prompt
Bunny Arabandi
Upgrade
Ask anything
Type here to search
33°C Sunny
ENG
10:28
15-07-2026
Ask Gemini
Upgrade